AERO-GUARD MISSION CONTROL AUDIT LEDGER

Autonomous Spacecraft Health Management (FDIR) • Closed-Loop Telemetry & Operator Audit Trail

MET STATUS: ACTIVE CRUISE
ORBIT: LEO 552 KM • INC 97.6°
HARDWARE: HITL PYSERIAL BRIDGE

LOG BUFFER: 3038 / 10000 LINES (FIFO ROLLING)

1. EXECUTIVE AUDIT METRICS & BUFFER TELEMETRY

TOTAL LOGS RECORDED	OPERATOR ACTIONS	EARLY PREDICTIONS	ANOMALIES DETECTED	RL MITIGATIONS	RECOVERIES
3038	35	1061	43	43	14
Capacity: 30.4%	Manual & Gates	Precursor Warnings	Multi-Class FDIR	Q-Learning Policy	Nominal Restored

2. SPACECRAFT SUBSYSTEM HEALTH & 5-FAULT COMPLIANCE ARCHITECTURE

Subsystem Component	Target Spacecraft Fault	Detection Precursor / Trigger	Closed-Loop Mitigation Policy	Recovery Verification Standard
1. PCDU Bus-A Power Distribution	**Internal Micro-Short** (Dendrite penetration)	Current > 4.5A spike Bus Voltage < 2.60V	**SAFETY_OVERRIDE / SAFE_MODE** Latching relay trips (<20ms); cross-straps to Bus-B	NASA-HDBK-4008 §4.2 OBC isolated at 9.2W
2. Li-ion 8S4P Pack Secondary Energy	**Thermal Runaway** (Joule heating accumulation)	dT/dt > 1.2°C/s Cell Temp T > 50°C	**LOAD_SHEDDING (-35% Payload)** RPC trips; BCR to Trickle; ADCS slews +18.5° to 3K sink	ECSS-E-ST-20C Joule heat drops 85%
3. BCR & Shunt Reg Solar Ingestion	**Deep Undervoltage** (Extended eclipse excursion)	Voltage < 2.90V SOC < 15%	**LOAD_SHEDDING (Tier-2 UVLS)** -75% Load shed; ADCS B-dot detumble slews arrays to Sun	ISRO-URSC-PCDU-EPS-04 Bus restores >3.3V
4. Li-ion Electrodes Electrochemical Cells	**High Impedance / Aging** (SEI Layer thickening)	R_int > 0.120 Ω Transient ESR ripple	**HIGH_SENSITIVITY_PREARM** RL lowers Tau=0.35; caps max discharge rate to 0.4C (1.5A)	NASA-HDBK-4008 Voltage sag stabilized
5. Sensor ADC Transducer Telemetry Bus	**Sensor Glitch / Noise** (Radiation SET transient)	> 3-sigma Z-Score outlier 12Hz ADC ringing	**AI RAG REASONER + EWMA** EWMA rejects spike; AI raises Tau=0.70 to prevent false trip	False Alarm Rate < 1.2% Payload stays online

3. CHRONOLOGICAL MISSION AUDIT LEDGER (UP TO 10,000-LINE BUFFER)

Full audit trail of all frontend user actions, streaming telemetry, early precursor predictions, anomaly triggers, RL mitigation actuations, and recovery events:

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#3038 T+00:35:59	INFO	**TELEMETRY FRAME**	EPS CORE	`V: 3.92V • I: 2.97A • T: 21.2°C • SOC: 98% • P_ens: 0.00`	Telemetry Cruise: V=3.919V | I=2.973A | T=21.2°C | SOC=97.7% | P_ens=0.001
#3037 T+00:35:56	INFO	**TELEMETRY FRAME**	EPS CORE	`V: 3.99V • I: 2.90A • T: 20.9°C • SOC: 98% • P_ens: 0.00`	Telemetry Cruise: V=3.988V | I=2.905A | T=20.9°C | SOC=97.7% | P_ens=0.001
#3036 T+00:35:56	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	`V: 3.99V • I: 2.90A • T: 20.9°C • SOC: 98% • P_ens: 0.00`	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#3035 T+00:35:54	INFO	**TELEMETRY FRAME**	EPS CORE	`V: 3.94V • I: 2.92A • T: 21.9°C • SOC: 98% • P_ens: 0.00`	Telemetry Cruise: V=3.935V | I=2.923A | T=21.9°C | SOC=97.8% | P_ens=0.001
#3034 T+00:35:53	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	`V: 3.91V • I: 2.90A • T: 21.3°C • SOC: 98% • P_ens: 0.00`	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#3033 T+00:35:48	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.93V • I: 2.93A • T: 21.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.926V | I=2.934A | T=21.5°C | SOC=97.8% | P_ens=0.001
#3032 T+00:35:47	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: OPEN_MISSION_AUDIT_CONSOLE. Opened 10k FIFO Mission Audit Ledger
#3031 T+00:35:47	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: CLEAR_ALL_FAULTS. Mitigated fault SENSOR_FAULT via LOAD_SHEDDING
#3030 T+00:35:47	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: CLEAR_ALL_FAULTS. All fault bus lines cleared, EPS telemetry reset to nominal.
#3029 T+00:35:47	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: LOAD_SHEDDING successfully mitigated SENSOR_FAULT. Subsystems normalized to nominal operating limits (V=3.93V, T=21.1°C).
#3028 T+00:35:46	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: CLEAR_ALL_FAULTS. Mitigated fault SENSOR_FAULT via LOAD_SHEDDING
#3027 T+00:35:46	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: CLEAR_ALL_FAULTS. All fault bus lines cleared, EPS telemetry reset to nominal.
#3026 T+00:35:46	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [SENSOR_FAULT]. Projected 60s Temp: 26.3°C, Safety Score: 0.71
#3025 T+00:35:46	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.30V • I: 4.57A • T: 21.3°C • SOC: 98% • P_ens: 0.98	Anomaly Triggered: SENSOR_FAULT | Severity: CRITICAL. P(Anomaly)=0.984 >= Tau=0.45. Status: ML_RL_ANOMALY_ALERT
#3024 T+00:35:45	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25s
#3023 T+00:35:45	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25.0 seconds.
#3022 T+00:35:44	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.98V • I: 2.90A • T: 21.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.984V | I=2.901A | T=21.9°C | SOC=97.8% | P_ens=0.001
#3021 T+00:35:42	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.93V • I: 2.96A • T: 21.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.930V | I=2.964A | T=21.1°C | SOC=97.9% | P_ens=0.002
#3020 T+00:35:42	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 3.93V • I: 2.96A • T: 21.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.002 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#3019 T+00:35:39	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.90V • I: 2.79A • T: 21.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.905V | I=2.787A | T=21.6°C | SOC=97.9% | P_ens=0.003
#3018 T+00:35:37	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.96V • I: 2.80A • T: 21.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.964V | I=2.796A | T=21.8°C | SOC=97.9% | P_ens=0.003
#3017 T+00:35:35	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.93V • I: 2.86A • T: 22.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.932V | I=2.863A | T=22.0°C | SOC=97.9% | P_ens=0.004
#3016 T+00:35:34	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.01V • I: 2.71A • T: 22.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.005 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#3015 T+00:35:32	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.95V • I: 2.70A • T: 21.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=3.953V | I=2.704A | T=21.7°C | SOC=98.0% | P_ens=0.003

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#3014 T+00:35:31	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: NOMINAL_MONITOR successfully mitigated NOMINAL. Subsystems normalized to nominal operating limits (V=3.96V, T=21.6°C).
#3013 T+00:35:31	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 3.96V • I: 2.73A • T: 21.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.003 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#3012 T+00:35:30	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.97V • I: 2.62A • T: 22.3°C	Deterministic Safety Guardrail Active! Violations: RAPID THERMAL EXCURSION: |dT/dt| = 13.29°C/s > 2.5°C/s
#3011 T+00:35:30	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 3.97V • I: 2.62A • T: 22.3°C • SOC: 98% • P_ens: 0.04	Anomaly Triggered: NOMINAL | Severity: EMERGENCY. P(Anomaly)=0.043 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#3010 T+00:35:30	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.80A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.219V | I=1.798A | T=35.6°C | SOC=98.0% | P_ens=0.001
#3009 T+00:35:27	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.72A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.168V | I=1.722A | T=36.2°C | SOC=98.0% | P_ens=0.001
#3008 T+00:35:25	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.78A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.202V | I=1.776A | T=35.2°C | SOC=98.0% | P_ens=0.001
#3007 T+00:35:24	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.76A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#3006 T+00:35:23	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.83A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#3005 T+00:35:22	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.71A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.185V | I=1.706A | T=35.9°C | SOC=98.0% | P_ens=0.001
#3004 T+00:35:20	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.80A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.190V | I=1.799A | T=35.7°C | SOC=98.0% | P_ens=0.001
#3003 T+00:35:17	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.79A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#3002 T+00:35:17	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.78A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.183V | I=1.781A | T=35.9°C | SOC=98.0% | P_ens=0.001
#3001 T+00:35:14	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.14V • I: 1.95A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.144V | I=1.953A | T=35.7°C | SOC=98.0% | P_ens=0.001
#3000 T+00:35:12	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.81A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.195V | I=1.806A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2999 T+00:35:12	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.81A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2998 T+00:35:10	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.73A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.192V | I=1.733A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2997 T+00:35:10	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.73A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2996 T+00:35:07	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.83A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.192V | I=1.829A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2995 T+00:35:05	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.79A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2994 T+00:35:05	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.14V • I: 1.83A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.140V | I=1.831A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2993 T+00:35:04	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.79A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2992 T+00:35:02	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.74A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.175V | I=1.739A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2991 T+00:35:01	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.83A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2990 T+00:35:00	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.83A • T: 36.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.150V | I=1.832A | T=36.3°C | SOC=98.0% | P_ens=0.000
#2989 T+00:34:57	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.86A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.150V | I=1.856A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2988 T+00:34:56	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.78A • T: 36.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2987 T+00:34:54	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.92A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.158V | I=1.917A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2986 T+00:34:52	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.80A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.193V | I=1.795A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2985 T+00:34:51	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.83A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2984 T+00:34:49	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.81A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.190V | I=1.813A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2983 T+00:34:47	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.92A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.164V | I=1.921A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2982 T+00:34:43	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.80A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.182V | I=1.796A | T=36.1°C | SOC=98.0% | P_ens=0.001
#2981 T+00:34:41	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.85A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.152V | I=1.855A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2980 T+00:34:41	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.15V • I: 1.85A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2979 T+00:34:39	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.84A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.172V | I=1.842A | T=36.1°C | SOC=98.0% | P_ens=0.001
#2978 T+00:34:39	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.84A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2977 T+00:34:36	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.74A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.151V | I=1.744A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2976 T+00:34:34	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.80A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.195V | I=1.800A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2975 T+00:34:32	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.80A • T: 34.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2974 T+00:34:31	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.81A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.183V | I=1.811A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2973 T+00:34:29	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.79A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.149V | I=1.792A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2972 T+00:34:27	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.78A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.170V | I=1.778A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2971 T+00:34:26	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.78A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2970 T+00:34:24	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.81A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.208V | I=1.805A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2969 T+00:34:23	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.14V • I: 1.84A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2968 T+00:34:22	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.78A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.206V | I=1.776A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2967 T+00:34:20	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.74A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2966 T+00:34:19	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.79A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.214V | I=1.791A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2965 T+00:34:17	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.78A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.197V | I=1.777A | T=36.1°C | SOC=98.0% | P_ens=0.001
#2964 T+00:34:16	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.14V • I: 1.85A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2963 T+00:34:14	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.24V • I: 1.83A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.240V | I=1.832A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2962 T+00:34:12	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.16V • I: 1.72A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2961 T+00:34:11	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.85A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.847A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2960 T+00:34:08	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.79A • T: 34.9°C • SOC: 98% • P_ens: 0.01	Telemetry Cruise: V=4.171V | I=1.785A | T=34.9°C | SOC=98.0% | P_ens=0.013
#2959 T+00:34:08	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.16V • I: 1.82A • T: 36.2°C • SOC: 98% • P_ens: 0.18	Early Precursor Detected: Ensemble Precursor Drift | P(Anomaly)=0.180 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2958 T+00:34:06	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.76A • T: 36.3°C • SOC: 98% • P_ens: 0.20	Telemetry Cruise: V=4.152V | I=1.762A | T=36.3°C | SOC=98.0% | P_ens=0.197
#2957 T+00:34:04	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.71A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.003 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2956 T+00:34:03	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.74A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.201V | I=1.742A | T=35.7°C | SOC=98.0% | P_ens=0.002
#2955 T+00:34:00	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.25V • I: 1.78A • T: 36.0°C • SOC: 98% • P_ens: 0.06	Telemetry Cruise: V=4.247V | I=1.781A | T=36.0°C | SOC=98.0% | P_ens=0.061
#2954 T+00:34:00	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: HIGH_SENSITIVITY_PREARM successfully mitigated SENSOR_FAULT. Subsystems normalized to nominal operating limits (V=4.25V, T=36.0°C).
#2953 T+00:33:57	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.89A • T: 35.3°C • SOC: 98% • P_ens: 0.61	Telemetry Cruise: V=4.176V | I=1.892A | T=35.3°C | SOC=98.0% | P_ens=0.609
#2952 T+00:33:56	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [SENSOR_FAULT]. Projected 60s Temp: 38.5°C, Safety Score: 0.57
#2951 T+00:33:54	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.46V • I: 2.66A • T: 35.6°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.464V > 4.35V
#2950 T+00:33:53	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.82V • I: 1.98A • T: 35.9°C • SOC: 98% • P_ens: 0.76	Telemetry Cruise: V=4.823V | I=1.976A | T=35.9°C | SOC=98.0% | P_ens=0.757
#2949 T+00:33:53	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.82V • I: 1.98A • T: 35.9°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.823V > 4.35V
#2948 T+00:33:52	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.91V • I: 1.19A • T: 36.4°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.915V > 4.35V
#2947 T+00:33:52	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.19V • I: 0.53A • T: 35.5°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.189V > 4.35V
#2946 T+00:33:51	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.42V • I: 0.19A • T: 35.5°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.420V > 4.35V
#2945 T+00:33:50	INFO	**TELEMETRY FRAME**	EPS CORE	V: 5.34V • I: -0.07A • T: 35.3°C • SOC: 98% • P_ens: 0.94	Telemetry Cruise: V=5.343V | I=-0.067A | T=35.3°C | SOC=98.0% | P_ens=0.943
#2944 T+00:33:50	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.34V • I: -0.07A • T: 35.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.343V > 4.35V
#2943 T+00:33:49	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.29V • I: 0.26A • T: 35.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.295V > 4.35V
#2942 T+00:33:48	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.77V • I: 0.66A • T: 35.6°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.770V > 4.35V
#2941 T+00:33:46	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.89V • I: 2.10A • T: 35.4°C • SOC: 98% • P_ens: 0.95	Telemetry Cruise: V=3.893V | I=2.105A | T=35.4°C | SOC=98.0% | P_ens=0.952
#2940 T+00:33:46	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [SENSOR_FAULT]. Projected 60s Temp: 26.9°C, Safety Score: 0.71

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2939 T+00:33:45	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [TRIGGER_SAFE_MODE] targeting [SENSOR_FAULT]. Projected 60s Temp: 38.3°C, Safety Score: 0.58
#2938 T+00:33:44	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.67V • I: 3.56A • T: 35.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.669V < 2.7V
#2937 T+00:33:43	INFO	**TELEMETRY FRAME**	EPS CORE	V: 2.80V • I: 3.43A • T: 35.9°C • SOC: 98% • P_ens: 0.99	Telemetry Cruise: V=2.802V | I=3.434A | T=35.9°C | SOC=98.0% | P_ens=0.992
#2936 T+00:33:40	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [SENSOR_FAULT]. Projected 60s Temp: 26.1°C, Safety Score: 0.85
#2935 T+00:33:39	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 0.59A • T: 35.2°C • SOC: 98% • P_ens: 0.85	Telemetry Cruise: V=4.188V | I=0.593A | T=35.2°C | SOC=98.0% | P_ens=0.852
#2934 T+00:33:38	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.51V • I: 0.21A • T: 35.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.514V > 4.35V
#2933 T+00:33:38	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [SENSOR_FAULT]. Projected 60s Temp: 25.9°C, Safety Score: 0.93
#2932 T+00:33:38	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.59V • I: 0.03A • T: 36.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.594V > 4.35V
#2931 T+00:33:37	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.80V • I: 0.16A • T: 35.5°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.803V > 4.35V
#2930 T+00:33:36	INFO	**TELEMETRY FRAME**	EPS CORE	V: 5.21V • I: 0.29A • T: 35.8°C • SOC: 98% • P_ens: 0.93	Telemetry Cruise: V=5.211V | I=0.293A | T=35.8°C | SOC=98.0% | P_ens=0.933
#2929 T+00:33:36	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.21V • I: 0.29A • T: 35.8°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.211V > 4.35V
#2928 T+00:33:36	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [TRIGGER_SAFE_MODE] targeting [SENSOR_FAULT]. Projected 60s Temp: 26.0°C, Safety Score: 0.87
#2927 T+00:33:35	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.56V • I: 0.62A • T: 35.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.557V > 4.35V
#2926 T+00:33:35	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.23V • I: 1.19A • T: 35.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.234V > 4.35V
#2925 T+00:33:34	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.32V • I: 1.92A • T: 35.8°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.317V > 4.35V
#2924 T+00:33:34	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [SENSOR_FAULT]. Projected 60s Temp: 26.5°C, Safety Score: 0.76
#2923 T+00:33:33	INFO	**TELEMETRY FRAME**	EPS CORE	V: 5.00V • I: 2.66A • T: 35.8°C • SOC: 98% • P_ens: 0.98	Telemetry Cruise: V=5.005V | I=2.664A | T=35.8°C | SOC=98.0% | P_ens=0.984
#2922 T+00:33:33	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.00V • I: 2.66A • T: 35.8°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.005V > 4.35V
#2921 T+00:33:32	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.70V • I: 3.35A • T: 36.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.701V > 4.35V

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2920 T+00:33:32	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [SENSOR_FAULT]. Projected 60s Temp: 34.2°C, Safety Score: 0.63
#2919 T+00:33:32	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.70V • I: 3.35A • T: 36.0°C • SOC: 98% • P_ens: 0.99	Anomaly Triggered: SENSOR_FAULT | Severity: EMERGENCY. P(Anomaly)=0.990 >= Tau=0.45. Status: CRITICAL_SAFETY_OVERRIDE
#2918 T+00:33:31	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25s
#2917 T+00:33:31	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25.0 seconds.
#2916 T+00:33:31	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.61V • I: 3.55A • T: 36.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.606V > 4.35V
#2915 T+00:33:31	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.61V • I: 3.55A • T: 36.0°C • SOC: 98% • P_ens: 0.32	Anomaly Triggered: INTERNAL_SHORT | Severity: EMERGENCY. P(Anomaly)=0.323 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#2914 T+00:33:30	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25s
#2913 T+00:33:30	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25.0 seconds.
#2912 T+00:33:30	INFO	**TELEMETRY FRAME**	EPS CORE	V: 5.16V • I: 2.81A • T: 35.7°C • SOC: 98% • P_ens: 0.21	Telemetry Cruise: V=5.161V | I=2.811A | T=35.7°C | SOC=98.0% | P_ens=0.210
#2911 T+00:33:30	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 5.16V • I: 2.81A • T: 35.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 5.161V > 4.35V
#2910 T+00:33:29	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.95V • I: 3.29A • T: 35.5°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.947V > 4.35V
#2909 T+00:33:29	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.95V • I: 3.29A • T: 35.5°C • SOC: 98% • P_ens: 0.26	Anomaly Triggered: NOMINAL | Severity: EMERGENCY. P(Anomaly)=0.262 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#2908 T+00:33:28	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.53V • I: 3.53A • T: 35.2°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-VOLTAGE: 4.531V > 4.35V
#2907 T+00:33:28	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.53V • I: 3.53A • T: 35.2°C • SOC: 98% • P_ens: 0.36	Anomaly Triggered: INTERNAL_SHORT | Severity: EMERGENCY. P(Anomaly)=0.364 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#2906 T+00:33:27	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25s
#2905 T+00:33:27	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_SENSOR_FAULT. Duration: 25.0 seconds.
#2904 T+00:33:27	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.78A • T: 36.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.002 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2903 T+00:33:27	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.74A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.193V | I=1.738A | T=35.6°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2902 T+00:33:24	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.79A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2901 T+00:33:24	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.79A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.163V | I=1.792A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2900 T+00:33:21	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.78A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.205V | I=1.776A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2899 T+00:33:21	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.78A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2898 T+00:33:19	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: HIGH_SENSITIVITY_PREARM successfully mitigated NOMINAL. Subsystems normalized to nominal operating limits (V=4.22V, T=35.9°C).
#2897 T+00:33:18	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.71A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.149V | I=1.706A | T=35.6°C | SOC=98.0% | P_ens=0.003
#2896 T+00:33:18	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.15V • I: 1.71A • T: 35.6°C	Deterministic Safety Guardrail Active! Violations: RAPID THERMAL EXCURSION: |dT/dt| = 43.00°C/s > 2.5°C/s
#2895 T+00:33:18	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [NOMINAL]. Projected 60s Temp: 26.2°C, Safety Score: 0.81
#2894 T+00:33:18	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.15V • I: 1.71A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Anomaly Triggered: NOMINAL | Severity: EMERGENCY. P(Anomaly)=0.003 >= Tau=0.35. Status: CRITICAL_SAFETY_OVERRIDE
#2893 T+00:33:18	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.61V • I: 3.33A • T: 78.6°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 78.56°C > 65.0°C
#2892 T+00:33:17	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.66V • I: 3.28A • T: 77.2°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 77.23°C > 65.0°C
#2891 T+00:33:17	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [INTERNAL_SHORT]. Projected 60s Temp: 26.6°C, Safety Score: 0.28
#2890 T+00:33:16	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.68V • I: 3.18A • T: 76.5°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 76.46°C > 65.0°C
#2889 T+00:33:15	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.70V • I: 3.14A • T: 75.3°C • SOC: 98% • P_ens: 0.96	Telemetry Cruise: V=3.701V | I=3.139A | T=75.3°C | SOC=98.0% | P_ens=0.963
#2888 T+00:33:15	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.70V • I: 3.14A • T: 75.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 75.31°C > 65.0°C
#2887 T+00:33:15	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.71V • I: 3.16A • T: 74.9°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 74.88°C > 65.0°C
#2886 T+00:33:14	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.70V • I: 3.13A • T: 73.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 73.75°C > 65.0°C
#2885 T+00:33:13	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.71V • I: 3.10A • T: 73.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 73.01°C > 65.0°C

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2884 T+00:33:13	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.72V • I: 3.12A • T: 72.0°C • SOC: 98% • P_ens: 0.96	Telemetry Cruise: V=3.718V | I=3.124A | T=72.0°C | SOC=98.0% | P_ens=0.958
#2883 T+00:33:13	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.72V • I: 3.12A • T: 72.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 71.98°C > 65.0°C
#2882 T+00:33:12	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [INTERNAL_SHORT]. Projected 60s Temp: 26.5°C, Safety Score: 0.26
#2881 T+00:33:12	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.76V • I: 2.98A • T: 71.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 70.99°C > 65.0°C
#2880 T+00:33:12	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [TRIGGER_SAFE_MODE] targeting [INTERNAL_SHORT]. Projected 60s Temp: 26.5°C, Safety Score: 0.14
#2879 T+00:33:11	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.72V • I: 3.02A • T: 69.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 69.31°C > 65.0°C
#2878 T+00:33:10	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.77V • I: 3.01A • T: 68.9°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 68.91°C > 65.0°C
#2877 T+00:33:10	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.78V • I: 2.95A • T: 68.1°C • SOC: 98% • P_ens: 0.94	Telemetry Cruise: V=3.783V | I=2.952A | T=68.1°C | SOC=98.0% | P_ens=0.940
#2876 T+00:33:10	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.78V • I: 2.95A • T: 68.1°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 68.11°C > 65.0°C
#2875 T+00:33:10	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [INTERNAL_SHORT]. Projected 60s Temp: 26.4°C, Safety Score: 0.30
#2874 T+00:33:09	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.75V • I: 2.94A • T: 68.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 67.97°C > 65.0°C
#2873 T+00:33:09	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [INTERNAL_SHORT]. Projected 60s Temp: 26.4°C, Safety Score: 0.26
#2872 T+00:33:08	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.77V • I: 2.87A • T: 65.8°C	Deterministic Safety Guardrail Active! Violations: CRITICAL OVER-TEMPERATURE: 65.82°C > 65.0°C
#2871 T+00:33:08	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 3.77V • I: 2.87A • T: 65.8°C • SOC: 98% • P_ens: 0.95	Anomaly Triggered: INTERNAL_SHORT | Severity: EMERGENCY. P(Anomaly)=0.952 >= Tau=0.35. Status: CRITICAL_SAFETY_OVERRIDE
#2870 T+00:33:08	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [THERMAL_RUNAWAY]. Projected 60s Temp: 27.3°C, Safety Score: 0.32
#2869 T+00:33:06	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.83V • I: 2.80A • T: 64.2°C • SOC: 98% • P_ens: 0.69	Telemetry Cruise: V=3.830V | I=2.805A | T=64.2°C | SOC=98.0% | P_ens=0.686
#2868 T+00:33:03	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.92V • I: 2.65A • T: 60.0°C • SOC: 98% • P_ens: 0.68	Telemetry Cruise: V=3.917V | I=2.654A | T=60.0°C | SOC=98.0% | P_ens=0.676
#2867 T+00:33:00	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.96V • I: 2.65A • T: 55.5°C • SOC: 98% • P_ens: 0.66	Telemetry Cruise: V=3.963V | I=2.654A | T=55.5°C | SOC=98.0% | P_ens=0.656
#2866 T+00:32:58	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 3.96V • I: 2.45A • T: 53.3°C • SOC: 98% • P_ens: 0.60	Anomaly Triggered: THERMAL_RUNAWAY | Severity: CRITICAL. P(Anomaly)=0.603 >= Tau=0.70. Status: NOMINAL

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2865 T+00:32:56	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.98V • I: 2.39A • T: 51.6°C • SOC: 98% • P_ens: 0.18	Telemetry Cruise: V=3.978V | I=2.386A | T=51.6°C | SOC=98.0% | P_ens=0.179
#2864 T+00:32:56	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 3.99V • I: 2.34A • T: 50.8°C • SOC: 98% • P_ens: 0.07	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.067 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2863 T+00:32:54	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: NOMINAL_MONITOR successfully mitigated NOMINAL. Subsystems normalized to nominal operating limits (V=4.01V, T=49.1°C).
#2862 T+00:32:54	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.00V • I: 2.18A • T: 47.8°C • SOC: 98% • P_ens: 0.04	Telemetry Cruise: V=3.998V | I=2.176A | T=47.8°C | SOC=98.0% | P_ens=0.036
#2861 T+00:32:54	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.00V • I: 2.18A • T: 47.8°C	Deterministic Safety Guardrail Active! Violations: RAPID THERMAL EXCURSION: |dT/dt| = 10.83°C/s > 2.5°C/s
#2860 T+00:32:53	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.00V • I: 2.18A • T: 47.8°C • SOC: 98% • P_ens: 0.04	Anomaly Triggered: NOMINAL | Severity: EMERGENCY. P(Anomaly)=0.036 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#2859 T+00:32:53	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_THERMAL_RUNAWAY. Duration: 25s
#2858 T+00:32:53	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_THERMAL_RUNAWAY. Duration: 25.0 seconds.
#2857 T+00:32:51	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.90V • I: 2.55A • T: 56.4°C • SOC: 98% • P_ens: 0.18	Telemetry Cruise: V=3.899V | I=2.551A | T=56.4°C | SOC=98.0% | P_ens=0.179
#2856 T+00:32:48	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.97V • I: 2.32A • T: 52.7°C • SOC: 98% • P_ens: 0.19	Telemetry Cruise: V=3.971V | I=2.316A | T=52.7°C | SOC=98.0% | P_ens=0.194
#2855 T+00:32:46	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.01V • I: 2.22A • T: 49.5°C • SOC: 98% • P_ens: 0.05	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.053 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2854 T+00:32:45	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.00V • I: 2.29A • T: 48.4°C • SOC: 98% • P_ens: 0.06	Telemetry Cruise: V=4.002V | I=2.292A | T=48.4°C | SOC=98.0% | P_ens=0.056
#2853 T+00:32:45	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: NOMINAL_MONITOR successfully mitigated NOMINAL. Subsystems normalized to nominal operating limits (V=4.00V, T=48.4°C).
#2852 T+00:32:44	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.01V • I: 2.24A • T: 48.2°C	Deterministic Safety Guardrail Active! Violations: RAPID THERMAL EXCURSION: |dT/dt| = 13.45°C/s > 2.5°C/s
#2851 T+00:32:44	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.01V • I: 2.24A • T: 48.2°C • SOC: 98% • P_ens: 0.12	Anomaly Triggered: NOMINAL | Severity: EMERGENCY. P(Anomaly)=0.125 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#2850 T+00:32:44	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_THERMAL_RUNAWAY. Duration: 25s
#2849 T+00:32:44	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_THERMAL_RUNAWAY. Duration: 25.0 seconds.
#2848 T+00:32:44	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.00V • I: 1.88A • T: 34.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 1.998V < 2.7V

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2847 T+00:32:43	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.01V • I: 1.72A • T: 35.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.009V < 2.7V
#2846 T+00:32:42	INFO	**TELEMETRY FRAME**	EPS CORE	V: 2.05V • I: 1.86A • T: 35.4°C • SOC: 10% • P_ens: 0.99	Telemetry Cruise: V=2.053V | I=1.859A | T=35.4°C | SOC=9.5% | P_ens=0.987
#2845 T+00:32:42	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.05V • I: 1.86A • T: 35.4°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.053V < 2.7V
#2844 T+00:32:41	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.06V • I: 1.92A • T: 35.1°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.064V < 2.7V
#2843 T+00:32:40	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.09V • I: 1.79A • T: 35.0°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.094V < 2.7V
#2842 T+00:32:39	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.09V • I: 1.86A • T: 34.9°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.086V < 2.7V
#2841 T+00:32:38	INFO	**TELEMETRY FRAME**	EPS CORE	V: 2.10V • I: 1.82A • T: 35.1°C • SOC: 10% • P_ens: 0.99	Telemetry Cruise: V=2.102V | I=1.823A | T=35.1°C | SOC=10.2% | P_ens=0.986
#2840 T+00:32:38	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.10V • I: 1.82A • T: 35.1°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.102V < 2.7V
#2839 T+00:32:37	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.12V • I: 1.86A • T: 35.8°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.119V < 2.7V
#2838 T+00:32:37	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.16V • I: 1.86A • T: 35.6°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.159V < 2.7V
#2837 T+00:32:36	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.20V • I: 1.81A • T: 35.2°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.198V < 2.7V
#2836 T+00:32:35	INFO	**TELEMETRY FRAME**	EPS CORE	V: 2.19V • I: 1.82A • T: 35.4°C • SOC: 11% • P_ens: 0.99	Telemetry Cruise: V=2.190V | I=1.821A | T=35.4°C | SOC=10.8% | P_ens=0.985
#2835 T+00:32:35	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.19V • I: 1.82A • T: 35.4°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.190V < 2.7V
#2834 T+00:32:34	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.23V • I: 1.80A • T: 35.1°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.227V < 2.7V
#2833 T+00:32:33	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.27V • I: 1.75A • T: 35.6°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.268V < 2.7V
#2832 T+00:32:32	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.29V • I: 1.85A • T: 35.4°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.288V < 2.7V
#2831 T+00:32:32	INFO	**TELEMETRY FRAME**	EPS CORE	V: 2.31V • I: 1.85A • T: 35.7°C • SOC: 12% • P_ens: 0.99	Telemetry Cruise: V=2.308V | I=1.848A | T=35.7°C | SOC=11.6% | P_ens=0.987
#2830 T+00:32:32	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.31V • I: 1.85A • T: 35.7°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.308V < 2.7V
#2829 T+00:32:31	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.32V • I: 1.75A • T: 35.8°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.321V < 2.7V
#2828 T+00:32:30	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.33V • I: 1.79A • T: 35.6°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.333V < 2.7V

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2827 T+00:32:29	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_UNDERVOLTAGE. Duration: 25s
#2826 T+00:32:29	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_UNDERVOLTAGE. Duration: 25.0 seconds.
#2825 T+00:32:29	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.29V • I: 1.78A • T: 35.3°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.293V < 2.7V
#2824 T+00:32:28	INFO	**TELEMETRY FRAME**	EPS CORE	V: 2.37V • I: 1.85A • T: 35.4°C • SOC: 12% • P_ens: 0.82	Telemetry Cruise: V=2.367V | I=1.848A | T=35.4°C | SOC=12.0% | P_ens=0.822
#2823 T+00:32:28	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.37V • I: 1.85A • T: 35.4°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.367V < 2.7V
#2822 T+00:32:27	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_UNDERVOLTAGE. Duration: 25s
#2821 T+00:32:27	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_UNDERVOLTAGE. Duration: 25.0 seconds.
#2820 T+00:32:27	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 2.32V • I: 1.77A • T: 35.9°C	Deterministic Safety Guardrail Active! Violations: CRITICAL UNDER-VOLTAGE (DEEP DISCHARGE): 2.323V < 2.7V
#2819 T+00:32:27	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 2.32V • I: 1.77A • T: 35.9°C • SOC: 12% • P_ens: 0.82	Anomaly Triggered: HIGH_IMPEDANCE | Severity: EMERGENCY. P(Anomaly)=0.817 >= Tau=0.70. Status: CRITICAL_SAFETY_OVERRIDE
#2818 T+00:32:26	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_UNDERVOLTAGE. Duration: 25s
#2817 T+00:32:26	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_UNDERVOLTAGE. Duration: 25.0 seconds.
#2816 T+00:32:25	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.64A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.184V | I=1.640A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2815 T+00:32:22	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.81A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2814 T+00:32:22	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.82A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.196V | I=1.821A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2813 T+00:32:19	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.71A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.171V | I=1.711A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2812 T+00:32:18	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.74A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2811 T+00:32:16	RECOVER	**RECOVERY EVENT**	SPACECRAFT HEALTH	N/A	Nominal State Restored: HIGH_SENSITIVITY_PREARM successfully mitigated NOMINAL. Subsystems normalized to nominal operating limits (V=4.19V, T=35.5°C).
#2810 T+00:32:16	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.78A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2809 T+00:32:16	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.79A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.176V | I=1.786A | T=36.2°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2808 T+00:32:16	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 4.18V • I: 1.79A • T: 36.2°C	Deterministic Safety Guardrail Active! Violations: RAPID THERMAL EXCURSION: |dT/dt| = 16.14°C/s > 2.5°C/s
#2807 T+00:32:16	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [NOMINAL]. Projected 60s Temp: 26.2°C, Safety Score: 0.84
#2806 T+00:32:16	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 4.18V • I: 1.79A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Anomaly Triggered: NOMINAL | Severity: EMERGENCY. P(Anomaly)=0.001 >= Tau=0.35. Status: CRITICAL_SAFETY_OVERRIDE
#2805 T+00:32:12	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.12V • I: 6.65A • T: 50.6°C • SOC: 98% • P_ens: 1.00	Telemetry Cruise: V=3.116V | I=6.649A | T=50.6°C | SOC=98.0% | P_ens=0.999
#2804 T+00:32:09	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.05V • I: 6.60A • T: 50.1°C • SOC: 98% • P_ens: 1.00	Telemetry Cruise: V=3.054V | I=6.598A | T=50.1°C | SOC=98.0% | P_ens=0.999
#2803 T+00:32:06	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.13V • I: 6.76A • T: 49.6°C • SOC: 98% • P_ens: 0.98	Telemetry Cruise: V=3.129V | I=6.760A | T=49.6°C | SOC=98.0% | P_ens=0.980
#2802 T+00:32:04	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [INTERNAL_SHORT]. Projected 60s Temp: 31.3°C, Safety Score: 0.53
#2801 T+00:32:03	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.12V • I: 6.71A • T: 47.8°C • SOC: 98% • P_ens: 0.97	Telemetry Cruise: V=3.122V | I=6.714A | T=47.8°C | SOC=98.0% | P_ens=0.970
#2800 T+00:31:59	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.17V • I: 6.60A • T: 45.6°C • SOC: 98% • P_ens: 0.98	Telemetry Cruise: V=3.171V | I=6.602A | T=45.6°C | SOC=98.0% | P_ens=0.976
#2799 T+00:31:56	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.16V • I: 6.63A • T: 43.5°C • SOC: 98% • P_ens: 0.99	Telemetry Cruise: V=3.159V | I=6.632A | T=43.5°C | SOC=98.0% | P_ens=0.994
#2798 T+00:31:56	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [INTERNAL_SHORT]. Projected 60s Temp: 39.6°C, Safety Score: 0.47
#2797 T+00:31:54	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [TRIGGER_SAFE_MODE] targeting [INTERNAL_SHORT]. Projected 60s Temp: 35.2°C, Safety Score: 0.46
#2796 T+00:31:52	INFO	**TELEMETRY FRAME**	EPS CORE	V: 3.15V • I: 6.66A • T: 42.3°C • SOC: 98% • P_ens: 0.99	Telemetry Cruise: V=3.151V | I=6.661A | T=42.3°C | SOC=98.0% | P_ens=0.991
#2795 T+00:31:52	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [HIGH_SENSITIVITY_PREARM] targeting [INTERNAL_SHORT]. Projected 60s Temp: 31.6°C, Safety Score: 0.56
#2794 T+00:31:51	SAFETY	**SAFETY OVERRIDE**	SAFETY GATEWAY	V: 3.12V • I: 6.44A • T: 41.4°C	Deterministic Safety Guardrail Active! Violations: RAPID THERMAL EXCURSION: |dT/dt| = 5.26°C/s > 2.5°C/s
#2793 T+00:31:51	RL-POLICY	**RL MITIGATION**	PCDU RL ACTUATOR	N/A	RL Mitigation Executed: [LOAD_SHEDDING] targeting [INTERNAL_SHORT]. Projected 60s Temp: 32.3°C, Safety Score: 0.51
#2792 T+00:31:51	CRITICAL	**FAULT ANOMALY**	EPS CORE	V: 3.12V • I: 6.44A • T: 41.4°C • SOC: 98% • P_ens: 1.00	Anomaly Triggered: INTERNAL_SHORT | Severity: EMERGENCY. P(Anomaly)=0.996 >= Tau=0.45. Status: CRITICAL_SAFETY_OVERRIDE
#2791 T+00:31:50	ACTION	**OPERATOR ACTION**	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_INTERNAL_SHORT. Duration: 25s

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2790 T+00:31:50	ACTION	OPERATOR ACTION	OPERATOR CONSOLE	N/A	Operator [MissionCommander] executed: INJECT_FAULT_INTERNAL_SHORT. Duration: 25.0 seconds.
#2789 T+00:31:49	INFO	TELEMETRY FRAME	EPS CORE	V: 4.17V • I: 1.86A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.166V | I=1.864A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2788 T+00:31:49	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.17V • I: 1.86A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2787 T+00:31:46	INFO	TELEMETRY FRAME	EPS CORE	V: 4.16V • I: 1.74A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.157V | I=1.742A | T=36.2°C | SOC=98.0% | P_ens=0.000
#2786 T+00:31:46	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.16V • I: 1.74A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.000 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2785 T+00:31:43	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.82A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.181V | I=1.818A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2784 T+00:31:40	INFO	TELEMETRY FRAME	EPS CORE	V: 4.21V • I: 1.86A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.215V | I=1.858A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2783 T+00:31:39	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.18V • I: 1.69A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2782 T+00:31:38	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.18V • I: 1.67A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2781 T+00:31:37	INFO	TELEMETRY FRAME	EPS CORE	V: 4.22V • I: 1.87A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.216V | I=1.868A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2780 T+00:31:34	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.83A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.183V | I=1.835A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2779 T+00:31:33	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.19V • I: 1.84A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2778 T+00:31:31	INFO	TELEMETRY FRAME	EPS CORE	V: 4.21V • I: 1.83A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.210V | I=1.826A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2777 T+00:31:30	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.17V • I: 1.70A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2776 T+00:31:28	INFO	TELEMETRY FRAME	EPS CORE	V: 4.15V • I: 1.72A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.149V | I=1.724A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2775 T+00:31:28	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.15V • I: 1.72A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2774 T+00:31:25	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.77A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.180V | I=1.771A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2773 T+00:31:23	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.19V • I: 1.88A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2772 T+00:31:22	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.73A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.173V | I=1.732A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2771 T+00:31:19	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.80A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.218V | I=1.799A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2770 T+00:31:18	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.14V • I: 2.53A • T: 36.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2769 T+00:31:16	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 2.52A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.218V | I=2.524A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2768 T+00:31:15	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 2.43A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2767 T+00:31:13	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 2.50A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.155V | I=2.498A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2766 T+00:31:12	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 2.47A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2765 T+00:31:10	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.13V • I: 2.58A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.126V | I=2.584A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2764 T+00:31:10	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.13V • I: 2.58A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2763 T+00:31:08	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.74A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.181V | I=1.737A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2762 T+00:31:05	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.88A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.163V | I=1.883A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2761 T+00:31:03	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.82A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.188V | I=1.817A | T=36.1°C | SOC=98.0% | P_ens=0.001
#2760 T+00:31:01	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.82A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.187V | I=1.822A | T=34.8°C | SOC=98.0% | P_ens=0.001
#2759 T+00:31:01	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.82A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2758 T+00:30:59	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.15V • I: 1.79A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2757 T+00:30:58	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.80A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.173V | I=1.800A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2756 T+00:30:56	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.87A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2755 T+00:30:55	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.86A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.174V | I=1.861A | T=35.4°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2754 T+00:30:53	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.13V • I: 1.80A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2753 T+00:30:53	INFO	TELEMETRY FRAME	EPS CORE	V: 4.22V • I: 1.69A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.221V | I=1.693A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2752 T+00:30:50	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.79A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.178V | I=1.793A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2751 T+00:30:48	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.20V • I: 1.85A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2750 T+00:30:48	INFO	TELEMETRY FRAME	EPS CORE	V: 4.19V • I: 1.91A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.191V | I=1.915A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2749 T+00:30:46	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.17V • I: 1.86A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2748 T+00:30:45	INFO	TELEMETRY FRAME	EPS CORE	V: 4.15V • I: 1.77A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.151V | I=1.773A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2747 T+00:30:43	INFO	TELEMETRY FRAME	EPS CORE	V: 4.21V • I: 1.78A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.214V | I=1.783A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2746 T+00:30:42	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.20V • I: 1.84A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2745 T+00:30:40	INFO	TELEMETRY FRAME	EPS CORE	V: 4.15V • I: 1.81A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.151V | I=1.812A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2744 T+00:30:39	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.19V • I: 1.81A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2743 T+00:30:38	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.86A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.180V | I=1.856A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2742 T+00:30:38	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.18V • I: 1.86A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2741 T+00:30:36	INFO	TELEMETRY FRAME	EPS CORE	V: 4.19V • I: 1.81A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.189V | I=1.815A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2740 T+00:30:33	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.79A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.182V | I=1.787A | T=36.1°C | SOC=98.0% | P_ens=0.001
#2739 T+00:30:33	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.18V • I: 1.79A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2738 T+00:30:31	INFO	TELEMETRY FRAME	EPS CORE	V: 4.19V • I: 1.83A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.190V | I=1.833A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2737 T+00:30:28	INFO	TELEMETRY FRAME	EPS CORE	V: 4.17V • I: 1.84A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.169V | I=1.838A | T=35.3°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2736 T+00:30:28	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.20V • I: 1.79A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2735 T+00:30:26	INFO	TELEMETRY FRAME	EPS CORE	V: 4.19V • I: 1.78A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.187V | I=1.782A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2734 T+00:30:24	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.16V • I: 1.86A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2733 T+00:30:23	INFO	TELEMETRY FRAME	EPS CORE	V: 4.17V • I: 1.84A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.170V | I=1.841A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2732 T+00:30:20	INFO	TELEMETRY FRAME	EPS CORE	V: 4.17V • I: 1.69A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.168V | I=1.688A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2731 T+00:30:20	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.17V • I: 1.69A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2730 T+00:30:17	INFO	TELEMETRY FRAME	EPS CORE	V: 4.23V • I: 1.80A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.225V | I=1.802A | T=36.1°C | SOC=98.0% | P_ens=0.001
#2729 T+00:30:17	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.23V • I: 1.80A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2728 T+00:30:14	INFO	TELEMETRY FRAME	EPS CORE	V: 4.19V • I: 1.76A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.188V | I=1.755A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2727 T+00:30:13	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.19V • I: 1.78A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2726 T+00:30:11	INFO	TELEMETRY FRAME	EPS CORE	V: 4.22V • I: 1.85A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.217V | I=1.847A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2725 T+00:30:10	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.18V • I: 1.91A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2724 T+00:30:08	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.76A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.756A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2723 T+00:30:07	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.19V • I: 1.77A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2722 T+00:30:06	INFO	TELEMETRY FRAME	EPS CORE	V: 4.20V • I: 1.83A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.200V | I=1.828A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2721 T+00:30:03	INFO	TELEMETRY FRAME	EPS CORE	V: 4.19V • I: 1.80A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.187V | I=1.800A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2720 T+00:30:03	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.19V • I: 1.80A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2719 T+00:30:00	INFO	TELEMETRY FRAME	EPS CORE	V: 4.18V • I: 1.80A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.177V | I=1.802A | T=35.1°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2718 T+00:29:59	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.23V • I: 1.85A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2717 T+00:29:58	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.83A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.173V | I=1.834A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2716 T+00:29:57	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.78A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2715 T+00:29:55	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.80A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.189V | I=1.805A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2714 T+00:29:54	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.72A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2713 T+00:29:53	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.83A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.194V | I=1.828A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2712 T+00:29:51	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.77A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2711 T+00:29:51	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.77A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.157V | I=1.772A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2710 T+00:29:47	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.84A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.219V | I=1.838A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2709 T+00:29:45	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.79A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.192V | I=1.791A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2708 T+00:29:44	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.80A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2707 T+00:29:43	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.78A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.191V | I=1.781A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2706 T+00:29:40	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.80A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.205V | I=1.796A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2705 T+00:29:38	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.82A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.156V | I=1.825A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2704 T+00:29:35	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.82A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.189V | I=1.817A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2703 T+00:29:35	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.82A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2702 T+00:29:33	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.87A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.869A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2701 T+00:29:31	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.79A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.207V | I=1.785A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2700 T+00:29:29	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.13V • I: 1.68A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2699 T+00:29:28	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.73A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.192V | I=1.729A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2698 T+00:29:26	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.86A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2697 T+00:29:26	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.80A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.146V | I=1.795A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2696 T+00:29:24	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.24V • I: 1.91A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2695 T+00:29:23	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.83A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.827A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2694 T+00:29:21	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.12V • I: 1.82A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2693 T+00:29:21	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.76A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.170V | I=1.760A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2692 T+00:29:19	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.73A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2691 T+00:29:18	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.77A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.200V | I=1.770A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2690 T+00:29:15	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.82A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.180V | I=1.823A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2689 T+00:29:15	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.75A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2688 T+00:29:13	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.78A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.203V | I=1.784A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2687 T+00:29:12	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.86A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2686 T+00:29:11	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.78A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2685 T+00:29:10	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.81A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.218V | I=1.813A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2684 T+00:29:08	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.87A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.866A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2683 T+00:29:05	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.75A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.196V | I=1.750A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2682 T+00:29:05	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.75A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2681 T+00:29:03	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.81A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.196V | I=1.805A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2680 T+00:29:01	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.75A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.207V | I=1.745A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2679 T+00:29:01	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.75A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2678 T+00:28:58	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.78A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.157V | I=1.782A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2677 T+00:28:58	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.16V • I: 1.82A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2676 T+00:28:56	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.77A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.213V | I=1.772A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2675 T+00:28:55	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.75A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2674 T+00:28:53	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.72A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.197V | I=1.719A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2673 T+00:28:51	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.78A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.168V | I=1.779A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2672 T+00:28:50	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.24V • I: 1.84A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2671 T+00:28:49	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.71A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.188V | I=1.713A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2670 T+00:28:46	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.79A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.184V | I=1.794A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2669 T+00:28:46	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.82A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2668 T+00:28:43	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.75A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.202V | I=1.752A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2667 T+00:28:43	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.75A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2666 T+00:28:41	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.15V • I: 1.87A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2665 T+00:28:41	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.85A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.171V | I=1.846A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2664 T+00:28:40	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.78A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2663 T+00:28:38	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.76A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.183V | I=1.759A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2662 T+00:28:38	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.76A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2661 T+00:28:36	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.77A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.186V | I=1.773A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2660 T+00:28:36	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.77A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2659 T+00:28:34	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.80A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.211V | I=1.797A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2658 T+00:28:32	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.22V • I: 1.80A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2657 T+00:28:31	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.24V • I: 1.81A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.239V | I=1.810A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2656 T+00:28:30	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.88A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2655 T+00:28:29	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.75A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.218V | I=1.752A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2654 T+00:28:27	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.81A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2653 T+00:28:26	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.85A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.194V | I=1.850A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2652 T+00:28:24	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.73A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.194V | I=1.731A | T=36.0°C | SOC=98.0% | P_ens=0.001
#2651 T+00:28:24	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.73A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2650 T+00:28:21	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.83A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.208V | I=1.831A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2649 T+00:28:21	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.83A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2648 T+00:28:20	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.22V • I: 1.76A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2647 T+00:28:19	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.74A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.165V | I=1.739A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2646 T+00:28:18	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.76A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2645 T+00:28:17	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.86A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.215V | I=1.862A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2644 T+00:28:16	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.82A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2643 T+00:28:14	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.78A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.189V | I=1.782A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2642 T+00:28:12	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.77A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.205V | I=1.769A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2641 T+00:28:09	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.75A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.182V | I=1.753A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2640 T+00:28:06	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.82A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.175V | I=1.816A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2639 T+00:28:06	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.82A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2638 T+00:28:04	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.83A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.204V | I=1.826A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2637 T+00:28:02	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.84A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2636 T+00:28:02	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.89A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.170V | I=1.887A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2635 T+00:27:59	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.81A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.200V | I=1.806A | T=36.2°C | SOC=98.0% | P_ens=0.001
#2634 T+00:27:59	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.81A • T: 36.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2633 T+00:27:57	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.12V • I: 1.74A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2632 T+00:27:57	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.84A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.199V | I=1.837A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2631 T+00:27:56	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.85A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2630 T+00:27:54	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.23V • I: 1.77A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.228V | I=1.770A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2629 T+00:27:52	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.84A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.186V | I=1.843A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2628 T+00:27:52	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.84A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2627 T+00:27:49	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.80A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.212V | I=1.797A | T=35.4°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2626 T+00:27:48	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.16V • I: 1.74A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2625 T+00:27:47	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.80A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.180V | I=1.801A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2624 T+00:27:46	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.24V • I: 1.86A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2623 T+00:27:45	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.85A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.199V | I=1.850A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2622 T+00:27:43	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.79A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2621 T+00:27:42	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.83A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.215V | I=1.828A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2620 T+00:27:40	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.85A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.189V | I=1.850A | T=35.9°C | SOC=98.0% | P_ens=0.001
#2619 T+00:27:40	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.85A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2618 T+00:27:37	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.82A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.197V | I=1.819A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2617 T+00:27:34	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.76A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.223V | I=1.755A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2616 T+00:27:32	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.91A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.178V | I=1.907A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2615 T+00:27:32	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.91A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2614 T+00:27:29	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.85A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.183V | I=1.853A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2613 T+00:27:27	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.75A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.173V | I=1.752A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2612 T+00:27:25	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.74A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.170V | I=1.742A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2611 T+00:27:23	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.77A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2610 T+00:27:22	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.78A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.186V | I=1.778A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2609 T+00:27:22	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.16V • I: 1.88A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2608 T+00:27:20	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.88A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.881A | T=35.9°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2607 T+00:27:20	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.88A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2606 T+00:27:17	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.76A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.169V | I=1.765A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2605 T+00:27:16	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.80A • T: 36.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2604 T+00:27:15	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.13V • I: 1.79A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.126V | I=1.789A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2603 T+00:27:14	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.78A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2602 T+00:27:13	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.76A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.193V | I=1.761A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2601 T+00:27:12	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.81A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2600 T+00:27:10	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.77A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.177V | I=1.769A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2599 T+00:27:08	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.85A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.182V | I=1.854A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2598 T+00:27:05	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.80A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.202V | I=1.797A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2597 T+00:27:04	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.15V • I: 1.65A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2596 T+00:27:02	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.84A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.152V | I=1.843A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2595 T+00:27:00	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.78A • T: 34.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.212V | I=1.785A | T=34.9°C | SOC=98.0% | P_ens=0.001
#2594 T+00:27:00	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.21V • I: 1.78A • T: 34.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2593 T+00:26:58	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.84A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2592 T+00:26:58	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.84A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.179V | I=1.842A | T=35.8°C | SOC=98.0% | P_ens=0.001
#2591 T+00:26:57	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.17V • I: 1.72A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2590 T+00:26:55	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.84A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.181V | I=1.837A | T=35.1°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2589 T+00:26:53	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 2.53A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.171V | I=2.529A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2588 T+00:26:52	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 2.50A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2587 T+00:26:50	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 2.60A • T: 34.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.187V | I=2.597A | T=34.9°C | SOC=98.0% | P_ens=0.001
#2586 T+00:26:50	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 2.60A • T: 34.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2585 T+00:26:48	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.11V • I: 2.58A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.106V | I=2.578A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2584 T+00:26:47	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 2.52A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2583 T+00:26:46	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.88A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.173V | I=1.877A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2582 T+00:26:43	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.88A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.192V | I=1.880A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2581 T+00:26:41	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.88A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.195V | I=1.878A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2580 T+00:26:40	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.82A • T: 35.9°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2579 T+00:26:38	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.82A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.197V | I=1.819A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2578 T+00:26:38	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.15V • I: 1.82A • T: 34.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2577 T+00:26:37	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.81A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2576 T+00:26:36	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.76A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.212V | I=1.761A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2575 T+00:26:34	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.81A • T: 34.9°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.173V | I=1.811A | T=34.9°C | SOC=98.0% | P_ens=0.001
#2574 T+00:26:33	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.23V • I: 1.75A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2573 T+00:26:31	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.79A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2572 T+00:26:30	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.82A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.198V | I=1.817A | T=35.7°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2571 T+00:26:29	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.19V • I: 1.75A • T: 34.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2570 T+00:26:28	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.22V • I: 1.77A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.218V | I=1.768A | T=35.6°C | SOC=98.0% | P_ens=0.001
#2569 T+00:26:25	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.21V • I: 1.79A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.213V | I=1.792A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2568 T+00:26:23	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.13V • I: 1.80A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.128V | I=1.800A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2567 T+00:26:23	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.13V • I: 1.80A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2566 T+00:26:21	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.17V • I: 1.81A • T: 35.7°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.168V | I=1.806A | T=35.7°C | SOC=98.0% | P_ens=0.001
#2565 T+00:26:18	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.20V • I: 1.73A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.200V | I=1.728A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2564 T+00:26:16	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.14V • I: 1.85A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.142V | I=1.846A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2563 T+00:26:15	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.12V • I: 1.77A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2562 T+00:26:13	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.15V • I: 1.88A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.148V | I=1.877A | T=35.1°C | SOC=98.0% | P_ens=0.001
#2561 T+00:26:12	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.20V • I: 1.80A • T: 35.6°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2560 T+00:26:11	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.80A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.156V | I=1.796A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2559 T+00:26:09	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.16V • I: 1.71A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.155V | I=1.706A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2558 T+00:26:07	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.15V • I: 1.83A • T: 35.1°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2557 T+00:26:06	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.80A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.176V | I=1.804A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2556 T+00:26:04	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.80A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.192V | I=1.803A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2555 T+00:26:03	PREDICT	**EARLY PREDICTION**	PREDICTIVE FDIR	V: 4.18V • I: 1.87A • T: 34.7°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2554 T+00:26:02	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.18V • I: 1.83A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.181V | I=1.833A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2553 T+00:25:59	INFO	**TELEMETRY FRAME**	EPS CORE	V: 4.19V • I: 1.86A • T: 36.1°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.187V | I=1.857A | T=36.1°C | SOC=98.0% | P_ens=0.001

SEQ / MET	SEVERITY	CATEGORY	SUBSYSTEM	TELEMETRY SNAPSHOT	AI/RL & MESSAGE DETAILS
#2552 T+00:25:58	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.17V • I: 1.83A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2551 T+00:25:57	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.16V • I: 1.67A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2550 T+00:25:56	INFO	TELEMETRY FRAME	EPS CORE	V: 4.16V • I: 1.79A • T: 35.2°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.164V | I=1.792A | T=35.2°C | SOC=98.0% | P_ens=0.001
#2549 T+00:25:54	INFO	TELEMETRY FRAME	EPS CORE	V: 4.20V • I: 1.78A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.195V | I=1.778A | T=34.8°C | SOC=98.0% | P_ens=0.001
#2548 T+00:25:54	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.20V • I: 1.78A • T: 34.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2547 T+00:25:51	INFO	TELEMETRY FRAME	EPS CORE	V: 4.17V • I: 1.82A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.165V | I=1.819A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2546 T+00:25:49	INFO	TELEMETRY FRAME	EPS CORE	V: 4.22V • I: 1.78A • T: 35.3°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.219V | I=1.778A | T=35.3°C | SOC=98.0% | P_ens=0.001
#2545 T+00:25:47	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.20V • I: 1.73A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR
#2544 T+00:25:47	INFO	TELEMETRY FRAME	EPS CORE	V: 4.20V • I: 1.82A • T: 35.5°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.201V | I=1.824A | T=35.5°C | SOC=98.0% | P_ens=0.001
#2543 T+00:25:44	INFO	TELEMETRY FRAME	EPS CORE	V: 4.22V • I: 1.76A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.223V | I=1.764A | T=35.0°C | SOC=98.0% | P_ens=0.001
#2542 T+00:25:44	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.18V • I: 1.81A • T: 35.8°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2541 T+00:25:42	INFO	TELEMETRY FRAME	EPS CORE	V: 4.15V • I: 1.82A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Telemetry Cruise: V=4.154V | I=1.823A | T=35.4°C | SOC=98.0% | P_ens=0.001
#2540 T+00:25:42	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.15V • I: 1.82A • T: 35.4°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.35. Recommended Pre-Arm: HIGH_SENSITIVITY_PREARM
#2539 T+00:25:41	PREDICT	EARLY PREDICTION	PREDICTIVE FDIR	V: 4.21V • I: 1.79A • T: 35.0°C • SOC: 98% • P_ens: 0.00	Early Precursor Detected: Thermal Gradient Rising | P(Anomaly)=0.001 >= Tau=0.70. Recommended Pre-Arm: NOMINAL_MONITOR